

- Q.17 Explain how can building orientation, window placement, and shading devices contribute to energy savings.
- Q.18 Describe how do site conditions and climate play a role in designing energy-efficient buildings.
- Q.19 Write short note on compliance documents required in BEE.
- Q.20 Describe strategies to minimize water consumption in buildings, including rainwater harvesting and efficient plumbing fixtures.
- Q.21 Explain the core principles of sustainable architecture in terms of energy efficiency.
- Q.22 Explain the steps involved in pursuing a green building certification.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Describe how to building certifications like LEED or BREEAM assess sustainable design practices.
- Q.24 Describe the passive design principles like orientation, natural ventilation, and daylighting be incorporated into sustainable architecture.
- Q.25 Explain the benefits of incorporating green roofs living walls into a design.

No. of Printed Pages : 4
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220264C

6th Sem / Architectural, Architectural (For Speech and Hearing Impaired)

Subject : Sustainable Architecture

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Which of the following is a key component of a green building's site design for water conservation ?
- a) Implementing permeable surfaces
 - b) Disregarding irrigation systems
 - c) Ignoring stormwater management
 - d) Installing water-intensive landscaping
- Q.2 What role does thermal mass play in green building design?
- a) Increasing heating and cooling loads
 - b) Storing and releasing heat to moderate temperature fluctuations
 - c) Ignoring indoor comfort levels
 - d) Minimizing energy efficiency

- Q.3 How does green building contribute to social sustainability?
- a) Ignoring community engagement
 - b) Focusing solely on individual benefits
 - c) Disregarding accessibility considerations
 - d) Promoting inclusivity, health, and well-being
- Q.4 Which sustainable practice helps reduce the urban heat island effect in a building?
- a) Increasing the density of heat-absorbing materials
 - b) Installing dark-colored pavements
 - c) Disregarding shade elements
 - d) Incorporating green spaces and trees
- Q.5 What does the term “Carbon footprint” refer to in the context of green buildings?
- a) The building’s age
 - b) The weight of the building
 - c) The building’s color
 - d) The amount of carbon dioxide emitted
- Q.6 What does LEED stand for in the context of green building certification?
- a) Low-efficiency Energy Design
 - b) Leadership in Energy and Environment Design
 - c) Leading Environmental Efficiency Development
 - d) Long-lasting Environmental Energy Deployment

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Sustainable architecture aims to minimize the _____ impact of buildings.
- Q.8 A building’s location and _____ can impact its energy consumption.
- Q.9 _____ is a process that involves converting waste materials into new products, reducing the need for virgin materials and landfill space.
- Q.10 Energy-efficient buildings can lower utility bills and reduce maintenance _____.
- Q.11 Sustainable design should consider the entire _____ of a building, from construction to demolition
- Q.12 Using renewable energy sources like solar or wind power in a building contributes to _____ efficiency.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the steps involved in pursuing a green building certification.
- Q.14 How can life cycle assessment be used to evaluate the sustainability of various building materials.
- Q.15 What criteria should be used to select sustainable building materials.
- Q.16 What strategies can be employed to minimize construction waste during the building process.